MCS Saturday SAT Math

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Digital SAT Math · Systems of Linear Equations

Systems Reasoning · Homework

New Digital Format · Algebra · Activity Sheet

Name _____ **Date** _____

Completion Check Show a valid setup, solve efficiently, and verify each correction.

A Read the System

6 prompts · 10 min

Do not solve unless asked. Read structure first: slope, intercept, coefficients, and equation relationships.

A1 · Classification Reason

The equations **3x + 2y = 14** and **6x + 4y = 28** form what kind of system? State the number of solutions and the structural evidence.

A2 · Parameter Analyze

For what value of **k** will **kx + 6y = 15** and **4x + 8y = 20** represent the same line? Explain using a common scale factor.

A3 · Coordinate Meaning Explain

A system has solution **(7, −2)**. Write the two substitution checks that must both be true for this ordered pair to solve the system.

A4 · Fast Recognition Audit

Without graphing, explain why **y = −3x + 5** and **6x + 2y = 4** cannot intersect.

A5 · Target Quantity Translate

A question asks for **twice the x-coordinate** of the intersection. If the system solution is **(a,b)**, what is the required response? Explain why **b** is irrelevant.

A6 · Representation Connect

Write a second equation that, with **2x − 5y = 9**, creates a system with no solution. Your equation may not be identical to the first.

B Choose the Cleanest Door

4 decisions · 7 min

Select the most efficient first move, then give one sentence of justification.

B1 Level 2

$y = 5x - 9$ and **$3x + 2y = 19$**

A. Substitution
B. Elimination
C. Graphing
D. Guess and check

Why: _____

B2 Level 2

$7x - 4y = 1$ and **$7x + 3y = 29$**

A. Substitution
B. Elimination
C. Graphing
D. Table

Why: _____

B3 Level 2

Two equations contain decimals and the prompt asks for the approximate intersection shown in a graphing context.

A. Substitution
B. Elimination
C. Desmos intersection
D. Mental math

Why: _____

B4 Level 2

The coefficients of **x** are 5 and 15, while no variable is isolated.

A. Isolate y first
B. Multiply one equation, then eliminate x
C. Graph by hand
D. Average coefficients

Why: _____

C Build, Solve, Verify

6 problems · 16 min

For each problem, show the key algebraic step. Check the ordered pair in both original equations.

C1 Level 1

What is the solution to the system **x + y = 11** and **x − y = 3**?

A. (4, 7) **B.** (7, 4)
C. (8, 3) **D.** (11, 3)

C2 Level 1

The system **y = 2x + 1** and **y = −x + 10** has solution **(x,y)**. What is **y**?

SPR: _____

C3 Level 2

What is the **x-coordinate** of the solution to **4x + 3y = 1** and **2x − 3y = 17**?

SPR: _____

C4 Level 2

For the system **5x + 2y = 16** and **3x − 2y = 8**, what is the value of **x + y**?

A. 2 **B.** 3
C. 4 **D.** 5

C5 Level 3

The graphs of **2x − y = 7** and **ax + y = 5** intersect at **(3,−1)**. What is the value of **a**?

SPR: _____

C6 Level 3

The solution to **3x + by = 17** and **x − y = 1** is **(5,4)**. What is **b**?

A. −1/2 **B.** 1/2
C. 2 **D.** 8

D Translation Lab

4 models · 12 min

Define both variables, write the system, solve, and answer the exact question with units.

D1 · Tickets Model

Adult tickets cost $18 and student tickets cost $11. A group buys 26 tickets for $349. How many student tickets were purchased?

D2 · Mixture Model

A lab needs 30 liters of a 42% solution by mixing a 30% solution with a 50% solution. How many liters of the 50% solution are needed?

D3 · Accounts Model

Two savings accounts contain $8,400 total. One earns 4% annually and the other 6%. Together they earn $438 in one year. How much is in the 6% account?

D4 · Data Plan Model

Plan A costs $24 plus $8 per gigabyte. Plan B costs $54 plus $3 per gigabyte. At how many gigabytes do the plans cost the same, and what is that common cost?

E Two-Pass Systems Set

10 questions · 18 min

Pass 1: collect direct points. Pass 2: return to parameter and modeling items. Leave no response blank.

Name the target | Exploit structure | Use Desmos deliberately | Verify in context

E1 Level 1

If **y = x + 6** and **y = 4x − 3**, what is the value of **x**?

A. 1
B. 2
C. 3
D. 4

E2 Level 1

What is the **y-coordinate** of the solution to **x + 2y = 13** and **x = 5**?

SPR: _____

E3 Level 2

Which ordered pair solves **2x + 5y = 4** and **x − y = 9**?

A. (7, −2)
B. (8, −1)
C. (9, −2)
D. (−7, 2)

E4 Level 2

The system **6x + 9y = 12** and **2x + 3y = k** has infinitely many solutions. What is **k**?

SPR: _____

E5 Level 2

A store sold 32 notebooks. Plain notebooks cost $4 and premium notebooks cost $7. The total revenue was $170. How many premium notebooks were sold?

A. 10
B. 12
C. 14
D. 18

E6 Level 3

Lines **ax + 4y = 12** and **3x − 2y = 5** are parallel. What is the value of **a**?

SPR: _____

E Two-Pass Systems Set · continued

10 questions · 18 min

Pass 1: collect direct points. Pass 2: return to parameter and modeling items. Leave no response blank.

Name the target | Exploit structure | Use Desmos deliberately | Verify in context

E7 Level 3

The solution to **px + y = 13** and **2x − y = 2** has x-coordinate 5. What is **p**?

A. 1/5 **B.** 6/5
C. 11/5 **D.** 3

E8 Level 3

For a positive constant **m**, the system **mx + 3y = 15** and **4x + 6y = 18** has no solution. What is **m**?

SPR: _____

E9 Level 4

A company sells small boxes for $12 and large boxes for $20. On one day it sold 75 boxes. Revenue from large boxes was $380 more than revenue from small boxes. How many large boxes were sold?

SPR: _____

E10 Level 4

In the system **(r+1)x + 2y = 8** and **6x + (r−1)y = 12**, there is no unique solution when the coefficient determinant is 0. What is the **positive** value of **r**?

A. √10 **B.** √11
C. √12 **D.** √13

F Process Evidence

After the timer · 6 min

Record observable evidence—not a general feeling about difficulty.

F1 Reflect

Which item did you flag first, and what exact feature made it expensive?

F2 Reflect

Where did elimination save steps compared with substitution?

F3 Reflect

If you used Desmos, what feature did you inspect? If not, where would it have helped?

F4 Reflect

Which answer did you verify in both equations or in context? Show that check.

H First Broken Line

4 forensic checks · 8 min

For any missed item, identify the earliest line at which the solution path stopped being valid.

H1 Audit

Problem number and first invalid line

H2 Audit

Why that line looked reasonable at the time

H3 Audit

A replacement line that preserves equivalence

H4 Audit

A fast check that would expose the error

I Error Log

Up to 6 corrections

Use one primary error type: Knowledge, Process, Reading, Timing, or Blank/Unreached.

#	Problem	My original thought	Primary error type	Correction rule
1				
2				
3				
4				
5				
6				

J Transfer Reflection

Complete sentences

The system feature I now notice before calculating is...

The cleanest method-switch I made was... because...

One parameter problem became simpler when I compared...

On my next Digital SAT module, I will verify a systems answer by...

G Bonus Challenge

3 Level 4 + 1 Level 5 · untimed

These are extension problems. Show a defensible path; a correct answer without supporting work is incomplete.

G1 Level 4

For real constants **a** and **b**, the system **ax + 3y = 7** and **4x + by = 10** has solution **(2,1)**. What is the value of **a+b**?

A. 2 **B.** 3
C. 4 **D.** 5

G2 Level 4

A theater sold 240 tickets. Balcony tickets cost $18 and orchestra tickets cost $30. After 20 balcony-ticket purchases were upgraded to orchestra tickets for an additional $12 each, total ticket revenue was $5,520. How many tickets were **originally** orchestra tickets?

SPR: _____

G3 Level 4

The system **(k−2)x + 3y = 9** and **6x + (k+1)y = 18** has infinitely many solutions. What is **k**?

SPR: _____

G4 Level 5

For constants **p** and **q**, every solution of **2x + py = 8** also satisfies **qx + 6y = 24**. The point **(1,t)** lies on both lines. What is the value of **p+q+t**?

SPR: _____

Bonus difficulty is intentionally above the core assignment.